

UBT516: BIOINFORMATICS

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to familiarize Students with basic concepts of sequences, structural alignment, database searching, protein structure prediction and applications.

Syllabus

Introduction: Goals, applications and limitations of Bioinformatics, Biological sequence and molecule file formats, Biological databases. Types of biological databases, its classification, application and pitfalls. Information retrieval from biological databases

Pairwise sequence alignment and database searching: Evolutionary Basis of sequence alignment, Homologous sequence, Global alignment and local alignment, Gap penalties, Scoring matrices, Methods of sequence alignment e.g., Dynamic programming methods: Needleman-Wunsch and Smith-Waterman algorithm, Database similarity search, Heuristic methods: FASTA and BLAST

Multiple sequence alignment and phylogenetics: Scoring multiple sequence alignments, Progressive alignment method, Iterative alignment method, Block-based alignment, Molecular evolution and phylogenetics, Phylogenetic analysis

Structural Bioinformatics: Protein structure basics, Visualization, Classification. Protein secondary structure prediction. Protein tertiary structure prediction: methods, applications, RNA structures, Genomics and proteomics. Machine learning in structure prediction

Laboratory Work: Biological database, file formats, Local and global sequence alignment of protein and DNA sequences, Phylogenetic tree construction, Sequence search and annotation

Course Learning Objectives (CLO)

The students will be able to:

1. Explore biological databases
2. Perform pairwise and multiple sequences alignment
3. Construct phylogenetic tree, analyze and make inferences from it.
4. Analyze sequence and structure of bio-macromolecule data.

Text Books

1. Xiong J, Essential Bioinformatics, Cambridge University Press (2013).

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

2. Mount D W, Bioinformatics - Sequence and Genome Analysis, Cold Spring Harbour Laboratory Press (2001), 2nd ed.
3. Ghosh Z, and Mallick B, Bioinformatics – Principles and Applications, Oxford University Press (2010).

Reference Books

1. Higgins, D. and Taylor, W., Bioinformatics: Sequence, Structure and Databanks- A Practical Approach, Oxford University Press (2000).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments /quizzes)	30

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